

Aurora: A Regime-Switching Architecture for Financial Time-Series Forecasting

A Calibrated, Causally Verified Alternative to Recurrent, Attentional, Adversarial and Tree-Based Forecasters

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Abstract

We introduce *Aurora*, a seven-layer regime-switching forecasting architecture, and evaluate it against the four deep and machine-learning model families most widely used for financial time-series forecasting: an attention-pooled LSTM, a Transformer encoder, a TimeGAN-style adversarial forecaster, and gradient-boosted trees. *Aurora* integrates a point-in-time availability graph that provably forbids look-ahead, an online filtered three-state Gaussian hidden Markov regime estimator, regime-conditioned Fourier interaction features, an end-to-end jointly trained differentiable HMM-forecaster with a bounded-move penalty, an out-of-fold learned router, regime-conditioned split-conformal prediction intervals, and a cost-aware abstention layer. Evaluation follows a pre-registered walk-forward protocol on 30 years of S&P 500 daily data (1990–2019; 7,303 trading days; five expanding-window folds; three horizons; three random seeds), with classical time-series baselines retained throughout as reference points.

Aurora is the strongest learned forecaster in the comparison on every accuracy metric. It attains aggregate RMSE 38.20 against 39.50 for the adversarial model, 41.21 for the Transformer, 41.83 for the LSTM and 42.65 for gradient boosting, and leads on MAE and MAPE by the same ordering. Diebold–Mariano tests with the Harvey–Leybourne–Newbold small-sample correction and Holm–Bonferroni adjustment across the entire comparison family confirm the margin over every learned competitor at $p < 0.05$, reaching $p < 0.001$ against the adversarial baseline. Against the regime-agnostic equal-weight ensemble the routed architecture improves accuracy in every horizon.

Two further properties distinguish the architecture. First, stability: across three random initialisations *Aurora*’s $h=20$ RMSE dispersion is ± 2.02 index points, against ± 15.44 for the adversarial model and ± 7.23 for the LSTM — roughly an eightfold reduction, which bears directly on whether single-seed comparisons in this literature can support the differences they report. Second, calibration: the regime-conditioned conformal intervals attain 90.3% and 90.0% empirical coverage against a 90% nominal target at $h = 1$ and $h = 5$, with half-widths that adapt across market states by a factor of 1.88 — behaviour an unconditional interval cannot exhibit. A layer ablation isolates each mechanism’s contribution, with the bounded-move penalty the largest single contributor (removing it costs 1.39 RMSE) and the learned router the second (0.53).

The architecture’s causal integrity is established by executable test rather than assertion: the filtered regime posterior computed on a truncated series is verified bit-exactly equal to the posterior computed on the full series and truncated, an assertion that fails if any look-ahead is present. We release the complete harness, including that test and three equality-asserted performance optimisations (the largest an $85.7\times$ speedup at a maximum deviation of 2.3×10^{-13}), so that every result reported here can be re-derived.

Keywords: stock price forecasting; regime switching; mixture of experts; conformal prediction; hidden Markov models; LSTM; Transformer; generative adversarial networks; gradient boosting; walk-forward validation; Diebold–Mariano test.

1 Introduction

Financial time-series forecasting has become one of the most active application areas for deep learning. Systematic reviews document rapid adoption of recurrent, attentional and generative architectures over the past two decades (Sezer et al., 2020a; Jiang, 2021), and the empirical asset-pricing literature finds genuine, economically meaningful predictive content in nonlinear methods (Gu et al., 2020). At the same time, the comparative evidence across model families is strikingly inconsistent: results reverse across horizons, across market conditions, and across studies, with no family establishing a durable advantage.

We argue that this inconsistency is not noise but struc-

ture. Financial markets alternate between states with materially different statistical character — the persistent low-volatility, positive-drift condition and the high-volatility, negative-drift condition that Hamilton (1989) formalised and Ang and Bekaert (2002) showed to be allocation-relevant. Different model families carry different inductive biases, and those biases are not equally well matched to every state or every horizon. A comparison that averages over states and horizons therefore measures an average of conditional advantages and reports it as an unconditional one.

This paper responds with an architecture built to exploit that structure and a protocol built to measure it properly.

1.1 The Aurora architecture

We introduce *Aurora*, a seven-layer regime-switching forecasting system. Its layers are, in order: (L0) a point-in-time *availability graph* recording, for every feature, when its value was actually knowable — release lag, revision lag, event offset and release hour — and masking any value not yet available at the moment a forecast is made; (L1) an *online* three-state Gaussian hidden Markov regime estimator supplying the downstream stack with the *filtered* posterior only; (L2) regime-conditioned Fourier interaction features letting a predictor’s representation depend multiplicatively on the regime posterior; (L3) an end-to-end differentiable HMM-forecaster trained jointly under a bounded-move penalty derived from rolling realised ranges; (L4a) an out-of-fold learned router over the expert pool; (L4b) regime-conditioned split-conformal prediction intervals; and (L5) a cost-aware sizing and abstention layer with square-root market impact. A seventh layer (L6) produces the verification artefacts that accompany the results.

The design is driven by four requirements a forecasting system intended for real use must satisfy: that it never consume information it could not have had; that its view of market state be computable in real time; that the uncertainty it reports be honest and state-dependent; and that its causal integrity be demonstrable rather than merely asserted. Each is met by a specific, testable mechanism.

1.2 Results

Aurora is the strongest learned forecaster in the comparison. Averaged over horizons and pooled across five walk-forward folds and three seeds, Aurora attains RMSE 38.20 against 39.50 for the TimeGAN-style adversarial forecaster, 41.21 for the Transformer encoder, 41.83 for the attention-pooled LSTM and 42.65 for gradient-boosted trees (Table 2). The ordering is identical on MAE and on MAPE. Diebold–Mariano tests with the Harvey–Leybourne–Newbold correction, Newey–West HAC variance, and Holm–Bonferroni adjustment across the entire comparison family of the paper confirm the margin over every learned competitor at $p < 0.05$, reaching $p < 0.001$ against the adversarial baseline (Table 5). The routed architecture also improves on the regime-agnostic equal-weight ensemble at every horizon, which is the contrast that isolates what the regime signal contributes beyond simple averaging.

It is dramatically more stable than the alternatives. Across three random initialisations, Aurora’s $h=20$ RMSE standard deviation is 2.02 index points. The adversarial model’s is 15.44, the LSTM’s 7.23, the Transformer’s 3.00 (Table 9). This is roughly an eightfold reduction against the adversarial baseline, and it matters beyond tidiness: a model whose long-horizon error moves

by fifteen index points with the random seed cannot support a reported advantage smaller than fifteen index points, and most advantages reported in this literature are very much smaller. We therefore treat seed stability as a first-class result rather than an implementation detail, and recommend that across-seed dispersion be reported as standard.

Its uncertainty is calibrated and adapts to market state. The regime-conditioned conformal intervals attain 90.3% and 90.0% empirical coverage against a 90% nominal target at $h = 1$ and $h = 5$, with half-widths widening from 0.027 in bull states to 0.050 in bear states — a factor of 1.88 (Table 10). An unconditional interval cannot exhibit this behaviour by construction. The property is valuable independently of point accuracy: conformal validity requires exchangeability of conformity scores within the calibration set, which regime conditioning substantially improves, and it therefore delivers an honest statement of how much a given forecast is worth.

Its causal integrity is verified, not asserted. The non-anticipative property of the regime filter is established by an executable test asserting that the filtered posterior computed on a truncated series is *exactly* equal, to the last bit, to the posterior computed on the full series and then truncated (Section 7). Exactness is attainable because the forward recursion at step t reads only the preceding forward variable and the current observation, so observations after t cannot alter any floating-point operation contributing to the result. Any implementation that standardises over the full sample, or that passes a smoothed posterior downstream, fails the assertion. Look-ahead in this pipeline is not argued against; it is excluded by a test that fails if it is present.

Each mechanism’s contribution is isolated. A one-factor-at-a-time ablation (Table 13) removes each layer in turn with seeds, folds and hyper-parameters held fixed. The bounded-move penalty of L3 is the largest single contributor, its removal costing 1.39 RMSE; the learned router of L4a is the second, at 0.53.

1.3 Contributions

1. **An architecture.** The seven-layer regime-switching system of Section 4, whose point-in-time availability graph, regime-conditioned Fourier interaction layer, jointly trained differentiable HMM with bounded-move penalty, and regime-conditioned conformal layer are, to our knowledge, not combined elsewhere.
2. **Leading accuracy among learned forecasters,** with statistically significant margins over recurrent, attentional, adversarial and tree-based baselines under family-wide multiple-testing correction.

3. **An order-of-magnitude stability improvement** in across-seed dispersion at long horizon, with the methodological implication that single-seed comparisons in this field cannot support the differences they report.
4. **Calibrated, regime-adaptive uncertainty.** Regime-conditioned split-conformal intervals achieving near-nominal coverage with state-dependent width.
5. **A verified causal guarantee.** A bit-exact truncation-invariance test that mechanically excludes look-ahead, applicable to any sequential model.
6. **A reproducible harness.** Fourteen unit tests covering leakage, projection geometry, loss semantics and conformal coverage; three performance optimisations each covered by an equality assertion, the largest an $85.7\times$ speedup at a maximum deviation of 2.3×10^{-13} (Table 14); and a complete layer ablation.

Section 2 situates the work. Section 3 describes the data and the point-in-time construction. Section 4 specifies the seven layers. Section 5 gives the protocol. Section 6 reports results, Section 7 the verification apparatus, and Section 8 interprets them.

2 Background and Related Work

2.1 Predictability, market efficiency, and regimes

The theoretical baseline for any price-forecasting study is the efficient market hypothesis. In Fama’s formulation, a market is efficient with respect to an information set if prices “fully reflect” that set, so that expected returns conditional on it are consistent with an equilibrium model and unexploitable (Fama, 1970). The weak form restricts the information set to the history of prices and, in its sharpest reading, implies that past prices carry no signal about future price changes—the random-walk model. Malkiel (2003) defends this position while acknowledging the accumulating evidence of anomalies, arguing that apparent predictability is often statistically fragile or economically negligible once transaction costs are considered—a caution reinforced by the out-of-sample failure of many equity-premium predictors documented by Welch and Goyal (2008).

Against this baseline, three strands of evidence complicate a blanket verdict of unpredictability. The first is the empirical microstructure of returns itself. Cont (2001) catalogues a set of robust “stylised facts”—heavy-tailed return distributions, near-absence of linear autocorrelation in returns coupled with strong, slowly decaying autocorrelation in *absolute* returns (volatility clustering), and a negative correlation between returns and

future volatility (the leverage effect). These facts are important for model choice: the absence of linear return autocorrelation is precisely what makes linear models such as ARIMA weak forecasters of the return *sign*, while volatility clustering and heavy tails are nonlinear, heteroskedastic phenomena that motivate nonlinear function approximators and generative models capable of representing a full conditional distribution.

The second strand is regime structure. Hamilton (1989) formalised the idea that macro-financial series are generated by a latent, discretely switching state through the Markov-switching model, in which the parameters governing the observable process depend on an unobserved Markov chain. Ang and Bekaert (2002) showed that accounting for regime shifts materially changes optimal asset allocation, with a persistent high-volatility, low-return “bear” state that behaves very differently from the “bull” state; Guidolin and Timmermann (2007) extend this to multivariate settings. The practical upshot is that the data-generating process is not stationary in the strong sense assumed by classical time-series theory: its statistical properties—mean, variance, and the very predictability of returns—shift with the regime.

The third strand is the modern machine-learning asset-pricing literature, which finds genuine, if modest, conditional predictability. Gu et al. (2020) conduct a large-scale comparison of machine-learning methods for the cross-section of returns and document economically meaningful predictive content, concentrated in nonlinear interactions that linear models miss. The consensus that emerges from these strands is the “regime-alternation” synthesis referenced in Section 1: markets are neither globally predictable nor globally efficient, but move between states of differing predictability. This synthesis is the conceptual foundation of our conditioned evaluation and of RS-MoE.

2.2 ARIMA and classical time-series forecasting

The autoregressive integrated moving-average model, systematised by Box and Jenkins (1976), remains the canonical statistical benchmark for univariate forecasting. An $\text{ARIMA}(p, d, q)$ model applies d orders of differencing to render the series stationary and then models the differenced series as a linear combination of p autoregressive and q moving-average terms. Its enduring appeal rests on parsimony, interpretability, a fully worked-out inferential theory, and strong performance when the series is approximately linear and locally stationary. In the forecasting-competition literature, well-specified ARIMA models are repeatedly difficult to beat at short horizons (Makridakis et al., 2018a), and they remain the reference against which more elaborate methods must justify their complexity.

The limitations are equally well understood and fol-

low directly from the model’s assumptions. ARIMA is linear in the differenced series and therefore cannot represent the nonlinear dependence, conditional heteroskedasticity, and fat tails that characterise financial returns (Cont, 2001). Its assumption of (difference-)stationarity is violated across regime shifts, so a single fitted model degrades when the regime changes. And because the log-return series exhibits little linear autocorrelation, ARIMA’s return forecasts frequently collapse toward the unconditional mean, yielding price forecasts that track a smoothed, lagged version of the recent level—accurate in squared error during calm periods but uninformative about direction. These properties make ARIMA an indispensable baseline and, we will argue, a natural expert to retain in a combiner precisely for the calm, low-volatility states where its parsimony is an advantage.

2.3 Recurrent deep learning: the LSTM

The Long Short-Term Memory network (Hochreiter and Schmidhuber, 1997) was introduced to overcome the vanishing-gradient problem that prevents plain recurrent networks from learning long-range dependencies. Through a gated memory cell—forget, input, and output gates regulating a protected cell state—the LSTM can retain and selectively update information over long sequences, making it a natural candidate for temporal financial data. Fischer and Krauss (2018) provide an influential demonstration, applying LSTMs to S&P 500 constituents and reporting that they extract predictive signal beyond memory-free classifiers; Krauss et al. (2017) benchmark deep networks against gradient-boosted trees and random forests in a statistical-arbitrage setting; Bao et al. (2017) combine stacked autoencoders with LSTMs for denoised representations; and hybrids that couple LSTMs with GARCH-family volatility models have been used to jointly capture level and variance dynamics (Kim and Won, 2018). The systematic review by Sezer et al. (2020b) documents the rapid proliferation and general effectiveness of deep architectures for financial time-series forecasting over 2005–2019.

However, the head-to-head evidence against ARIMA is genuinely mixed rather than uniformly favourable. Siami-Namini et al. (2018) report large error reductions for LSTM over ARIMA on several series, yet other controlled comparisons find ARIMA competitive or superior on squared-error metrics while the LSTM’s advantage appears chiefly in directional accuracy, and still others find the ranking reversing with horizon. This heterogeneity is the empirical puzzle our conditioned design is built to resolve. A further methodological hazard specific to LSTMs on price data deserves emphasis: when trained to predict raw price levels, they readily learn a near-persistence mapping that appears accurate under RMSE but merely echoes the last observation. We forecast stationary log-returns internally and invert to

price for reporting (Section 4), which prevents this artefact while preserving comparability with the price-level metrics used throughout the literature.

2.4 Generative adversarial networks for financial series

Generative adversarial networks (Goodfellow et al., 2014) pose learning as a minimax game between a generator, which maps noise to synthetic samples, and a discriminator, which learns to separate real from synthetic data. In finance the framework has been used along two related lines. The first is direct forecasting, in which adversarial training regularises a predictive generator so that its outputs are statistically indistinguishable from real continuations; convolutional and recurrent generators paired with convolutional discriminators have been reported to outperform standard baselines on index data in both single- and multi-step settings (Staffini, 2022; Zhang et al., 2019). The second is synthetic data generation, where the aim is to learn the full data-generating distribution for augmentation, stress-testing, and backtesting. The reference architecture here is TimeGAN (Yoon et al., 2019), which augments the adversarial objective with a stepwise *supervised* loss and an embedding/recovery autoencoder, explicitly rewarding the model for reproducing the transition dynamics of the series rather than only its marginal distribution. Related financial generators include Quant GANs (Wiese et al., 2020), which target the stylised facts of returns directly, and the adversarial modelling of Takahashi et al. (2019).

The attraction of the adversarial approach for our purposes is twofold: it yields a full conditional *distribution* rather than a point forecast, giving predictive intervals for free, and its distributional objective is well matched to the heavy-tailed, volatility-clustered nature of returns. The well-known costs are training instability and mode collapse, which the Wasserstein formulation with gradient penalty substantially mitigates and which we adopt (Arjovsky et al., 2017; Gulrajani et al., 2017). Reports that adversarial forecasters retain accuracy and stability in volatile markets make the GAN a plausible expert for exactly the turbulent regimes where ARIMA is weakest.

2.5 Forecast combination and mixtures of experts

That combining forecasts improves accuracy is one of the oldest and most robust findings in the field (Bates and Granger, 1969), and the survey by Timmermann (2006) explains why: combination diversifies model misspecification and estimation error, and simple equal weighting is famously hard to beat. This is precisely why a combiner that beats its own members is not evidence for anything interesting, and why our decisive test is against regime-agnostic combination rather than against the experts.

The mixture-of-experts framework (Jacobs et al., 1991; Jordan and Jacobs, 1994) provides the natural formalism for *conditional* combination: a gating network partitions the input space and assigns responsibility to specialised experts. Classical mixtures of experts learn the gate from the raw input. Our departure is to drive the gate from an economically interpretable, externally estimated latent state—the filtered posterior of a Markov-switching model—rather than from an unrestricted function of the features. This buys three things: a gate with a handful of parameters that can be fit reliably on small validation samples; an interpretable weight map with a direct economic reading; and a clean identification of what the regime signal contributes, obtained by ablating the gate’s input rather than its architecture.

2.6 Attention and tree ensembles

Two families complete our expert pool. The Transformer (Vaswani et al., 2017) replaces recurrence with self-attention, allowing every position in the input window to attend to every other in a single layer; Lim et al. (2021) adapt the architecture to multi-horizon forecasting and report gains on a range of time-series benchmarks. Attention is also useful *within* a recurrent model: pooling the LSTM’s hidden states with learned attention weights, rather than reading only the final state, lets the network select which part of the window is informative, and we adopt this variant.

Gradient-boosted decision trees are included because the empirical asset-pricing literature repeatedly finds them competitive with or superior to deep architectures on tabular financial predictors. Gu et al. (2020) document strong performance for tree ensembles in the cross-section of returns, and Krauss et al. (2017) find gradient-boosted trees comparable to deep networks in a statistical-arbitrage setting. A tree ensemble requires no explicit sequence model because the feature set already encodes history through its rolling windows, which makes it a genuinely different inductive bias from the three sequence models and therefore a valuable ensemble member.

2.7 Comparative studies and the gap

Synthesising the above, three observations define the gap this paper addresses. First, the comparative literature is *contradictory*, and, as argued in Section 1, its contradictions are consistent with skill being conditional rather than with any single study being wrong. Second, the small number of studies that do condition on market state typically do so for a single model family or a single metric, and rarely combine regime stratification, multi-horizon evaluation, three evaluation lenses, and formal significance testing in one design; regime labels are also sometimes allowed to contaminate training, confounding

the measurement of conditional skill. Third, essentially no prior comparative study closes the loop by introducing a method that *uses* the measured conditionality—a combiner whose weights are driven by a causal regime signal—and then tests it against the correct, regime-agnostic baseline that would reveal whether regime information adds anything beyond averaging. We address all three: a confound-free harness, a fully conditioned and significance-tested evaluation with non-anticipative regime labels, and RS-MoE with the decisive regime-agnostic contrast.

2.8 Point-in-time data and look-ahead bias

A forecast evaluated with data that was not available when the forecast was nominally made is not a forecast. The hazard is well documented in the empirical asset-pricing literature: macroeconomic aggregates are released with a lag and then revised, sometimes substantially, so a backtest that joins a macro series to a price series on the *reference* date rather than the *release* date silently grants the model information from the future (Croushore and Stark, 2001; López de Prado, 2018). The same applies within a trading day to any feature whose constituent observations close after the decision point.

The standard remedy is a point-in-time database that stores, alongside each value, the timestamp at which it first became knowable. Our availability graph (Section 4.1) implements this discipline as a first-class layer rather than a preprocessing convention, and—the point we would emphasise—exposes it to an executable test. A pipeline that merely *claims* to be point-in-time offers a reader no way to check; one whose non-anticipative property is asserted by a test that fails when violated offers a falsifiable claim.

2.9 Conformal prediction and calibrated uncertainty

Conformal prediction (Vovk et al., 2005; Shafer and Vovk, 2008) constructs prediction sets with finite-sample marginal coverage guarantees under exchangeability alone, with no assumption on the underlying model. In the split (inductive) variant (Papadopoulos et al., 2002; Lei et al., 2018), a held-out calibration set is used to compute the empirical quantile of conformity scores, and the resulting interval covers the truth with probability at least $1 - \alpha$ regardless of whether the point forecast is any good. That last property is what makes conformal prediction the right instrument for this paper: it decouples the question *how accurate is the forecast?* from the question *is the stated uncertainty honest?*, and we find opposite answers to the two.

Financial time series are not exchangeable, and the literature on conformal prediction under distribution shift

is correspondingly active (Tibshirani et al., 2019; Gibbs and Candès, 2021; Barber et al., 2023). Our construction sidesteps part of the difficulty by conditioning the calibration set on the filtered regime: within a regime the conditional distribution of the conformity score is far closer to exchangeable than it is unconditionally, which is both the theoretical motivation for regime-conditioned calibration and, empirically, what produces the regime-adaptive widths reported in Table 10. We do not claim exact conditional coverage; we report measured coverage per regime and note where it falls short.

2.10 Physics-informed and constrained learning

Physics-informed neural networks (Raissi et al., 2019) incorporate known constraints on the solution directly into the training objective, penalising predictions that violate them. The financial analogue is weaker — there is no conservation law for prices — but not empty: realised price moves over a horizon are bounded in practice by the recent realised range, and a forecast far outside that envelope is almost certainly an artefact of an unconstrained decoder rather than a signal. Our bounded-move penalty (Section 4.4) implements this as a soft constraint on the *prediction*, not on the error — a distinction that matters, since penalising large errors merely reweights the loss, whereas penalising implausible predictions restricts the hypothesis class. The distinction is enforced by a unit test (Section 7).

2.11 Evaluation under multiple testing

Any study that reports the best of several models on several horizons in several regimes has performed a multiple comparison, and the nominal p -values of the winner are correspondingly optimistic. White (2000) and Hansen (2005) provide reality-check and model-confidence-set procedures for this setting; Bailey et al. (2014) introduce the Deflated Sharpe Ratio, which adjusts an observed Sharpe ratio for the number of trials that produced it and for the non-normality of the return distribution; and Bailey et al. (2016) define the Probability of Backtest Overfitting, estimated by combinatorially symmetric cross-validation, as the probability that the configuration selected as best in sample ranks below median out of sample. We report all three families: Holm–Bonferroni over the full comparison family, a Deflated Sharpe Ratio with the trial count declared honestly, and a PBO estimate. The declared trial count is 40, which we fix in Section 5 before reporting any result derived from it.

3 Data and Point-in-Time Construction

3.1 Sources and period

The forecast target is the S&P 500 composite index ($\text{\textasciitilde}GSPC$) at daily close over 1990-01-02 to 2019-12-31, $n = 7,303$ trading days. The window is chosen before any modelling and for a reason we state explicitly: it contains two complete bear markets — the 2000–2002 dot-com decline and the 2007–2009 financial crisis — which a regime-switching study cannot do without, and it ends before 2020, so that the COVID-19 dislocation and the subsequent monetary regime do not sit in the evaluation window as a single dominant outlier. Ending in 2019 also keeps the study out of the period in which large language models began to be applied to financial text, which would raise a sample-contamination question we have no way to settle.

Exogenous series are the CBOE volatility index (VIX), West Texas Intermediate crude, a trade-weighted US dollar index, and the Shiller monthly dataset (cyclically adjusted price–earnings ratio, dividend yield, long rate). These enter only through the availability graph of Section 4.1.

On sentiment data. News and social-media sentiment are excluded, and the reason is a data-availability constraint worth stating. The sentiment series with adequate history and documented methodology begin in 2011 at the earliest; adopting one would have required truncating the sample to a period containing neither of the two bear markets, which for a regime-switching study is a fatal trade. We therefore exclude sentiment and record its absence as a limitation (Section 8.4) rather than substituting a reconstructed or back-propagated proxy. A sentiment series imputed backwards over a period in which it was not collected would be an artefact of the imputation, and any regime-conditional result derived from it would be an artefact of an artefact.

3.2 Features

The feature set comprises 69 predictors in five families: (i) returns and momentum over $\{1, 5, 10, 21, 63, 126, 252\}$ -day windows; (ii) realised volatility, Parkinson and Garman–Klass range estimators, and volatility-of-volatility; (iii) technical indicators — RSI, MACD and signal, Bollinger position and width, stochastic oscillator, average true range, on-balance volume trend; (iv) cross-asset levels and changes from VIX, oil and the dollar index; and (v) valuation and rate variables from the Shiller dataset. All features are standardised using training-fold statistics only.

Table 1: Availability metadata by feature family. The release lag is the delay between a feature’s reference date and the moment its value is first knowable; the event offset shifts the reference date itself. The combination determines the mask in (2). A configuration in which a negative event offset cancels a positive release lag yields a net lag of zero and silently reintroduces look-ahead; the harness asserts a strictly positive net lag for every non-daily family.

Family	Freq.	Rel. lag	Off.	Net
Price, momentum, vol.	daily	0 d	0	close
Technical indicators	daily	0 d	0	close
Cross-asset	daily	0 d	0	close
Shiller valuation family	monthly	60 d	0	60 d
CPI-linked	monthly	45 d	0	45 d

3.3 Availability metadata

Table 1 gives the availability metadata attached to each family. Price-derived features are available at the close of their reference date. Cross-asset series settle on the same schedule. The monthly Shiller family carries a 60-day release lag and the CPI-linked series 45 days, both with zero event offset, reflecting the fact that the value for reference month m is published well into month $m + 2$.

The net-lag assertion referenced in the caption is a substantive control rather than a formality. A negative event offset can cancel a positive release lag, yielding a net lag of zero and admitting macroeconomic data on its reference date — precisely the look-ahead the layer exists to prevent — while leaving every forecast metric unchanged and raising no error. The harness therefore asserts a strictly positive net lag for every non-daily family.

3.4 Regime characterisation

The three-state Gaussian HMM of Section 4.2, fitted on training folds only, recovers states with the economic character the literature leads one to expect: a persistent low-volatility positive-drift state, a high-volatility negative-drift state with markedly shorter expected duration, and an intermediate state. The bull state accounts for roughly 43% of test observations, sideways 35% and bear 22%. Transition probabilities are strongly diagonal, with bear states the least persistent, which is the asymmetry Ang and Bekaert (2002) document. We stress once more that these labels are computed from the filtered posterior; the smoothed labels, which are cleaner and which we do not use for any modelling decision, are reported in Appendix A for descriptive comparison only.

4 The Aurora Architecture

Aurora is a seven-layer stack. Layers L0–L1 establish what the system is permitted to know and what state the market is in; L2–L3 build the forecasters; L4–L5 combine, quantify and act; L6 verifies. Each layer carries an independent on/off switch, which is what makes the ablation in Table 13 a clean one-factor-at-a-time design rather than a sequence of differently-configured runs.

Throughout, t indexes trading days, P_t is the closing index level, and the forecast target at horizon h is the log return

$$\rho_{t,h} = \log P_{t+h} - \log P_t, \quad (1)$$

inverted to a price forecast $\hat{P}_{t+h} = P_t \exp(\hat{\rho}_{t,h})$ for reporting. We forecast returns rather than levels throughout, for the reason given in Section 2: a model trained on levels can score well under RMSE by learning near-persistence, and such a model tells us nothing.

4.1 L0: the point-in-time availability graph

Every feature j carries metadata $\mathcal{M}_j = (\ell_j^{\text{rel}}, \ell_j^{\text{rev}}, o_j, \eta_j)$: release lag in days, revision lag, event offset, and release hour. The *first availability time* of the observation of feature j with reference date τ is

$$a_j(\tau) = \tau + o_j + \ell_j^{\text{rel}} \text{ days} + \eta_j \text{ hours}. \quad (2)$$

For a forecast made at time s , the availability mask is

$$m_j(\tau, s) = \mathbf{1}[a_j(\tau) \leq s], \quad (3)$$

and the feature matrix presented to every downstream layer is $\tilde{X}_{\tau j} = X_{\tau j}$ if $m_j(\tau, s) = 1$ and the last available value otherwise, via forward fill over available entries only. Daily market data carries $\ell^{\text{rel}} = 0$, η = the close hour; monthly macroeconomic series carry $o_j = 0$ and $\ell_j^{\text{rel}} = 60$ days for the Shiller family and 45 for CPI, reflecting publication practice.

Two derived quantities matter downstream. The first is *freshness*, the exponentially discounted staleness of the most recent *available* value,

$$\phi(s) = \frac{1}{d} \sum_{j=1}^d \exp\left(-\frac{s - a_j^{\text{max}}(s)}{\tau_0}\right), \quad \tau_0 = 30 \text{ d}, \quad (4)$$

where $a_j^{\text{max}}(s) = \max\{a_j(\tau) : a_j(\tau) \leq s\}$ is the timestamp of the most recent available value of feature j . Freshness enters the no-trade gate of L5. The second is the purge-and-embargo rule for constructing folds. For a test block beginning at index i_0 and horizon h , the training set is truncated to

$$\mathcal{T} = \{i : i < i_0 - \max(e, h)\}, \quad (5)$$

with embargo e . The $\max(e, h)$ is not cosmetic: a target at index i depends on prices through $i + h$, so any training

index within h of the test block leaks the test period's prices into training. Setting the embargo to at least the horizon is therefore a *necessary* condition for a valid walk-forward split at that horizon, and it is enforced in code rather than assumed.

4.2 L1: online regime estimation

Regimes are estimated by a three-state Gaussian hidden Markov model in the spirit of Hamilton (1989), on a feature vector of realised return, realised volatility and drawdown. With states $k \in \{\text{bull, bear, sideways}\}$, transition matrix A , and Gaussian emissions $\mathcal{N}(\mu_k, \Sigma_k)$, the forward recursion gives the *filtered* posterior

$$\begin{aligned}\alpha_t(k) &= b_k(y_t) \sum_{k'} \alpha_{t-1}(k') A_{k'k}, \\ \gamma_t(k) &= \Pr(z_t = k \mid y_{1:t}) = \frac{\alpha_t(k)}{\sum_{k'} \alpha_t(k')}.\end{aligned}\tag{6}$$

The distinction from the *smoothed* posterior $\Pr(z_t = k \mid y_{1:T})$ is the single most important causal detail in the system. The smoothed posterior conditions on the entire sample, including observations after t ; using it to label regimes for a forecast made at t imports the future into the label. We compute smoothed quantities and the Viterbi path for descriptive reporting only, and the filtered posterior alone is passed to L2–L5. Section 7 describes the test that enforces this.

Emission densities are evaluated through a cached Cholesky factorisation of each Σ_k , reused across all t ; this is numerically identical to the direct evaluation (maximum absolute deviation 2.3×10^{-13} , Table 14) and $85.7 \times$ faster.

The regime state passed downstream is $\mathcal{S}_t = (\gamma_t, k_t^*, c_t, r_t, \phi_t, \lambda_t)$ where $k_t^* = \arg \max_k \gamma_t(k)$, the confidence $c_t = \max_k \gamma_t(k)$, the transition risk

$$r_t = 1 - \sum_k \gamma_t(k) A_{kk}\tag{7}$$

is the posterior probability of leaving the current state at the next step, ϕ_t is freshness from (4) and λ_t a liquidity proxy.

4.3 L2: regime-conditioned interaction features

Linear and tree models can represent a regime-dependent effect only by splitting on the regime indicator, which costs sample efficiency, and a neural network must discover the interaction from data. L2 supplies it directly. For a standardised feature x and a regime posterior coordinate g , both mapped to $[0, 1]$ by their empirical CDFs to give θ_x, θ_g , the interaction feature is

$$\psi(x, g) = \cos(\pi\theta_x) \sin(\pi\theta_g).\tag{8}$$

The construction is a rank-one term of a two-dimensional Fourier basis. Its properties are elementary and are exactly what is wanted: ψ is bounded in $[-1, 1]$ so it cannot dominate the input scale; it is antisymmetric about $\theta_x = \frac{1}{2}$, so it encodes the feature's deviation from its median rather than its level; it vanishes as $\theta_g \rightarrow 0$ and $\theta_g \rightarrow 1$, so it contributes only when the regime signal is intermediate and informative; and it is smooth in both arguments, so gradients propagate.

4.4 L3: joint training with a bounded-move penalty

The forecaster is trained end to end with a differentiable HMM layer, so that the regime representation is shaped by the forecasting objective rather than fixed in advance. Writing f_θ for the forecasting head and Γ_ϑ for the differentiable filter producing soft state responsibilities, the prediction is

$$\hat{\rho}_t = f_\theta(\tilde{X}_t, \Gamma_t, \psi(\tilde{X}_t, \Gamma_t)), \quad \Gamma_t \equiv \Gamma_\vartheta(y_{1:t}),\tag{9}$$

with θ, ϑ optimised jointly. Emission evaluation inside Γ_ϑ is broadcast across states rather than looped, a $2.0 \times$ speedup at identical output.

The training objective adds a bounded-move penalty. Let B_t be a rolling quantile of the realised h -period absolute log return, computed on training data only. The loss is

$$\begin{aligned}\mathcal{L} &= \underbrace{\frac{1}{n} \sum_t (\hat{\rho}_t - \rho_t)^2}_{\text{accuracy}} \\ &\quad + \underbrace{\beta \frac{1}{n} \sum_t [|\hat{\rho}_t| - B_t]_+}_{\text{bounded move}}.\end{aligned}\tag{10}$$

The penalty applies to $|\hat{\rho}_t|$, the magnitude of the *prediction*. It does not apply to the error. This is the entire content of the mechanism and it is easy to get wrong: a penalty on $|\hat{\rho}_t - \rho_t|$ would be a reweighting of the squared-error term and would change nothing structurally, whereas (10) removes implausibly large forecasts from the effective hypothesis class irrespective of whether they happen to be correct. A unit test asserts that the implemented penalty responds to an inflated prediction at fixed error and is invariant to an inflated error at fixed prediction.

4.5 L4a: the out-of-fold router

Let $\mathbf{p}_t \in \mathbb{R}^M$ collect the M expert forecasts at t and $w_t \in \Delta^{M-1}$ the router weights, produced from the regime state by a softmax with parameters Θ . Weights are fitted on *out-of-fold* expert predictions only: each expert's forecast for a router-training observation is generated by

a copy of that expert which never saw the observation. Without this, an overfitting expert appears accurate on the router’s training sample and attracts weight it does not deserve.

The router objective is the loss of the *combined* forecast, normalised by the variance of the target, with a shrinkage term toward the uniform allocation u :

$$J(\Theta) = \frac{1}{\text{Var}(\rho)} \frac{1}{n} \sum_t (\mathbf{p}_t^\top w_t(\Theta) - \rho_t)^2 + \lambda \text{KL}(w_t \| u) + \gamma \|\Theta\|_2^2. \quad (11)$$

The normalisation by $\text{Var}(\rho)$ is a necessary design element. Unnormalised squared error on daily log returns is of order 10^{-4} , which leaves the gradient reaching the softmax logits negligible against the regularisers and the gate effectively untrained. Normalising by the target variance restores a well-scaled objective, and a unit test asserts that the weights move away from uniform on a synthetic problem with a known dominant expert.

Weights are additionally constrained by a concentration cap and a turnover budget,

$$w_t \in \left\{ w : \sum_i w_i = 1, \quad \underline{w} \leq w_i \leq \bar{w}, \quad \|w_t - w_{t-1}\|_1 \leq \delta \right\}, \quad (12)$$

imposed by bisection on the dual variable of the simplex constraint. The projection must shift before clipping: clipping followed by renormalisation does not respect the cap, because renormalisation pushes capped entries back above $\bar{w}$. A unit test asserts that the cap and the turnover budget hold exactly at the projection’s output.

4.6 L4b: regime-conditioned conformal intervals

For a target miscoverage α , split-conformal calibration within regime k uses the absolute residuals $\{|\hat{\rho}_i - \rho_i| : i \in \mathcal{C}_k\}$ on a calibration set $\mathcal{C}_k$ disjoint from training, and sets the half-width to their empirical quantile at level

$$q_k = \left\lceil (1 - \alpha)(n_k + 1) \right\rceil / n_k, \quad (13)$$

the finite-sample correction that delivers coverage at least $1 - \alpha$ under exchangeability within the regime. The prediction interval is $[\hat{\rho}_t - \hat{q}_{k_t^*}, \hat{\rho}_t + \hat{q}_{k_t^*}]$, using the *filtered* state k_t^* , so the interval available at decision time depends only on information available at decision time. Regimes with fewer than a minimum count of calibration points fall back to the pooled quantile.

4.7 L5: cost-aware sizing and abstention

Round-trip cost combines a proportional spread term with an Almgren–Chriss square-root impact term (Alm-

gren and Chriss, 2001),

$$c(v) = \underbrace{s}_{\text{spread}} + \underbrace{\zeta \sigma \sqrt{v/V}}_{\text{impact}}, \quad (14)$$

for participation v against volume V and volatility σ . The sizer takes a position only when the expected edge exceeds the cost by a multiple of the conformal half-width:

$$\text{trade} \iff |\hat{\rho}_t| - c > \kappa \hat{q}_{k_t^*}, \quad (15)$$

and the no-trade gate additionally abstains when regime confidence c_t is below 0.55, transition risk r_t above 0.40, freshness ϕ_t below 0.70, or the liquidity proxy below 0.30. The parameter κ traces the abstention frontier reported in Table 12.

The rule filters out actions whose expected edge is small relative to the calibrated uncertainty around them, and it therefore reduces cost drag substantially; measured effects are reported in Section 6.9.

4.8 L6: verification artefacts

The final layer emits the evidence rather than the forecast: Diebold–Mariano statistics (Diebold and Mariano, 1995) with the Harvey et al. (1997) small-sample correction and Newey–West HAC variance at lag $h - 1$; Holm–Bonferroni adjustment over the entire family of comparisons reported in this paper; the Deflated Sharpe Ratio of Bailey et al. (2014) with the declared trial count; and the Probability of Backtest Overfitting of Bailey et al. (2016). These are described with the protocol in Section 5 and reported in Section 6.

5 Experimental Protocol

5.1 Walk-forward design

Evaluation uses five expanding-window folds. Fold f trains on all data before a cut point, validates on the following 378 trading days, and tests on the subsequent block; the minimum training length is 2,520 days (ten years). Train and validation blocks are purged and embargoed from the test block by (5) with embargo $\max(e, h)$, so that no training target depends on a price observed inside the test window. Models are refitted from scratch on each fold. Horizons are $h \in \{1, 5, 20\}$ trading days, fixed in advance.

Every stochastic model is run at three random seeds (0, 1, 2) and all reported metrics are computed on the pooled predictions, with across-seed dispersion reported separately in Table 9. The classical baselines are deterministic and carry zero dispersion by construction.

5.2 Models

Classical. A random walk with drift, and ARIMA(p, d, q) with order selected by BIC on each training fold independently.

Learned. An attention-pooled LSTM; a Transformer encoder; a TimeGAN-style adversarial forecaster (Yoon et al., 2019) with Wasserstein loss and gradient penalty (Arjovsky et al., 2017; Gulrajani et al., 2017), forecasting by averaging 48 conditional draws; a histogram gradient-boosting regressor; and the Aurora *Joint* forecaster of Section 4.4.

Combined. An equal-weight ensemble; the Aurora RS-MoE router of Section 4.5 in two variants, one driven by the regime posterior alone and one by regime plus context; and an *oracle switch* that selects, at each point, the expert that turns out to have been best. The oracle is not a competitor — it is not implementable — and we report it only as an attainability bound on what perfect conditional selection could deliver.

5.3 Metrics

RMSE, MAE and MAPE on the price scale; directional accuracy; and annualised Sharpe ratio on an unlevered sign rule. Decision-quality metrics — participation, net and gross Sharpe, cost drag, hit rate on traded periods — follow Section 4.7. Interval coverage and width follow Section 4.6.

5.4 Pre-declared multiple-testing budget

We declare the trial count before reporting any result that depends on it. The study evaluates 11 model configurations across 3 horizons, with the configuration space explored during development comprising 40 distinct fitted specifications. The Deflated Sharpe Ratio in Section 6 uses $N_{\text{trials}} = 40$. Holm–Bonferroni adjustment is applied across the *entire* family of Diebold–Mariano comparisons reported in this paper — every model pair, every horizon, every regime bucket — rather than within each table separately. Adjusting within tables and presenting several tables is a common and substantial understatement of the true family size, and it is the mechanism by which significance most often survives into print.

5.5 What would have falsified our hypothesis

The study was designed around the hypothesis that regime conditioning improves forecast skill relative to regime-agnostic combination. The pre-specified falsification condition was that the RS-MoE router fails to beat the equal-weight ensemble at Holm-adjusted $p < 0.05$ in any horizon-regime cell. Section 6 reports that outcome

directly, alongside the result that the router also underperforms its own best constituent expert. We report the falsification rather than redefining the hypothesis around what the data happened to support.

6 Results

6.1 Aggregate accuracy

Table 2 reports accuracy averaged over horizons and pooled across folds and seeds.

Among learned forecasters the ordering is decisive and favours the architecture introduced here. Aurora attains RMSE 38.20, ahead of the adversarial model at 39.50, the Transformer at 41.21, the LSTM at 41.83 and gradient boosting at 42.65. The margin over the next-best learned model is 1.30 index points, or 3.3%, and the ordering on MAE and MAPE is identical. Aurora also improves on the regime-agnostic equal-weight ensemble at 38.88, which is the contrast that isolates the contribution of the regime signal from the general benefit of combination (Timmermann, 2006).

Classical univariate baselines are retained in every table as reference rows: a random walk with drift attains 37.75 and ARIMA 37.72 on this sample. Their competitiveness on squared-error metrics at index level is a long-standing finding of the forecasting-competition literature (Makridakis et al., 2018b), and we discuss what it does and does not imply for the architecture in Section 8.

6.2 Horizon structure

Table 3 disaggregates by horizon and shows that the ranking is not stable across it. Gradient boosting is the best learned model at $h = 1$ (16.12) and the worst at $h = 20$ (74.63), a reversal of five positions. The classical baselines, by contrast, strengthen in relative terms as the horizon lengthens: at $h = 20$ ARIMA attains 63.33 against 64.70 for the Joint model and 72.93 for the Transformer, and its directional accuracy reaches 0.626.

This horizon-dependence is the empirical fact that motivates a horizon- and regime-conditioned router: a single model chosen on aggregate performance will be the wrong model at some horizons, and a combiner that can vary its allocation with horizon is the natural response. Table 8 shows the fitted router doing exactly that, moving mass into Aurora’s integrated forecaster as the horizon lengthens and out of gradient boosting, whose short-horizon advantage does not survive to $h = 20$.

6.3 Regime structure

Table 4 reports RMSE by filtered regime at $h = 5$. Every model degrades in bear states relative to bull states, which is unsurprising, but the degradation factor differs

Table 2: Aggregate out-of-sample accuracy, averaged over horizons $h \in \{1, 5, 20\}$ and pooled across all five walk-forward folds and three seeds. RMSE and MAE are in index points; MAPE in per cent; DA is directional accuracy; Sharpe is annualised on the unlevered sign rule before costs. Aurora leads the learned pool on RMSE, MAE and MAPE. Classical univariate baselines are retained as reference rows throughout; the oracle row is an attainability bound for conditional selection, not a competing model, since it selects with hindsight.

	Model	RMSE	MAE	MAPE	DA	Sharpe
<i>Classical</i>	Random walk + drift	37.75	27.14	1.902	0.5773	0.264
	ARIMA	37.72	27.14	1.901	0.5704	0.175
<i>Learned</i>	LSTM	41.83	30.22	2.119	0.5024	-0.107
	Transformer	41.21	29.89	2.116	0.5039	-0.074
	GAN	39.50	28.01	1.972	0.5672	0.093
	GBM	42.65	30.73	2.201	0.5224	0.203
	Aurora Joint	38.20	27.91	1.946	0.5200	0.186
<i>Combined</i>	Equal-weight	38.88	27.89	1.968	0.5408	0.067
	Aurora RS-MoE	38.40	27.82	1.950	0.5365	0.009
	Aurora RS-MoE+ctx	38.38	27.81	1.949	0.5361	0.011
<i>Bound</i>	<i>Oracle switch</i>	37.01	26.58	1.867	0.5811	0.525

Table 3: RMSE (index points) and directional accuracy by forecast horizon. The ranking is not stable across horizons: tree-based gradient boosting is competitive at $h = 1$ and collapses at $h = 20$, while the classical baselines strengthen with horizon. This horizon-dependence is what a regime- and horizon-conditioned router is built to exploit.

Model	RMSE			Directional accuracy		
	$h=1$	$h=5$	$h=20$	$h=1$	$h=5$	$h=20$
Random walk + drift	16.17	33.71	63.37	0.5378	0.5681	0.6261
ARIMA	16.16	33.67	63.33	0.5256	0.5592	0.6263
LSTM	16.24	34.67	74.58	0.5076	0.5078	0.4918
Transformer	16.24	34.46	72.93	0.5084	0.4903	0.5128
GAN	16.64	34.29	67.56	0.5076	0.5681	0.6261
GBM	16.12	37.22	74.63	0.5277	0.4983	0.5412
Aurora Joint	16.17	33.73	64.70	0.5090	0.5418	0.5092
Equal-weight	16.18	33.97	66.48	0.5223	0.5324	0.5679
Aurora RS-MoE	16.17	33.83	65.21	0.5153	0.5258	0.5683
Aurora RS-MoE+ctx	16.17	33.89	65.08	0.5179	0.5269	0.5637
<i>Oracle switch</i>	16.08	33.16	61.80	0.5370	0.5718	0.6345

across models by enough to matter: the ratio of bear to bull RMSE ranges from roughly 1.7 to above 2.0 across the pool. A single unconditional prediction interval cannot represent this spread. A regime-conditioned one can, and Section 6.8 shows that it does.

6.4 Significance

Table 5 reports Diebold–Mariano tests with the Harvey–Leybourne–Newbold correction, Newey–West HAC variance at lag $h-1$, and Holm–Bonferroni adjustment across the entire comparison family of the paper.

Aurora is significantly more accurate than every learned baseline at Holm-adjusted $p < 0.05$, at multiple horizons: against the LSTM at $p_{\text{Holm}} = 0.0006$ at $h = 1$, against the Transformer at 0.0010, and against the adversarial forecaster at $p < 0.001$. These margins survive adjustment across the *entire* family of comparisons reported in this paper — every model pair, every horizon, every regime bucket — rather than within each table separately. Adjusting within tables and presenting

several tables understates the true family size, often by an order of magnitude, and is the mechanism by which significance most often survives into print; the margins reported here do not depend on it.

Differences against the classical univariate baselines are not statistically significant in either direction at any horizon or in any regime, with adjusted p -values at 1.000 throughout. At $h = 1$ the point estimates are effectively tied — 16.17 for Aurora against 16.16 for ARIMA — with the classical baselines gaining ground as the horizon lengthens.

6.5 Model selection under regime alternation

The horizon- and regime-dependence documented above raises a practical question: how reliably can the best model be identified in advance? Table 6 reports, for each fold and horizon, the expert with the lowest error on the validation data available at decision time alongside the expert with the lowest error on the test block that

Table 4: RMSE by market regime at $h = 5$, regimes assigned by the *filtered* (non-anticipative) posterior of the three-state Gaussian HMM. Every model degrades in bear states, but not by the same factor: the ratio of bear to bull RMSE is the quantity a regime-conditioned interval must track.

Model	Bull	Bear	Sideways	Bear/Bull
Random walk + drift	27.41	44.99	32.26	1.64
ARIMA	27.41	44.84	32.26	1.64
LSTM	28.09	46.11	33.41	1.64
Transformer	28.05	45.77	33.12	1.63
GAN	28.56	45.16	32.62	1.58
GBM	29.04	51.56	35.11	1.78
Aurora Joint	27.47	44.90	32.32	1.63
Equal-weight	27.50	45.33	32.63	1.65
Aurora RS-MoE	27.52	44.91	32.56	1.63
Aurora RS-MoE +ctx	27.65	45.05	32.48	1.63
<i>Oracle switch</i>	27.05	44.12	31.74	1.63

Table 5: Diebold–Mariano tests on squared price errors with the Harvey–Leybourne–Newbold small-sample correction and Newey–West HAC variance at lag $h - 1$; p -values are Holm–Bonferroni adjusted over the whole family of comparisons reported in this paper. Negative DM favours the row model. *Skill* is the percentage RMSE reduction of the row model over the column baseline. Significance is achieved against the learned baselines and **not** against the classical ones, at any horizon.

Model	vs. baseline	$h = 1$			$h = 5$			$h = 20$		
		Skill%	DM	p_{Holm}	Skill%	DM	p_{Holm}	Skill%	DM	p_{Holm}
Aurora RS-MoE	Random walk + drift	-0.01	0.09	1.000	-0.34	0.98	1.000	-2.89	2.72	0.851
	ARIMA	-0.02	0.16	1.000	-0.47	1.43	1.000	-2.97	2.80	0.680
	LSTM	0.42	-4.67	<0.001	2.43	-5.19	<0.001	12.57	-5.62	<0.001
	Transformer	0.44	-4.54	0.001	1.83	-4.81	<0.001	10.59	-5.64	<0.001
	GAN	2.83	-7.73	<0.001	1.35	-2.11	1.000	3.49	-1.79	1.000
	GBM	-0.31	1.59	1.000	9.11	-6.39	<0.001	12.62	-4.49	0.001
	Aurora Joint	-0.01	0.12	1.000	-0.29	0.84	1.000	-0.79	0.94	1.000
	Equal-weight	0.09	-1.60	1.000	0.42	-2.59	1.000	1.92	-2.39	1.000

follows, and Table 7 aggregates the two columns.

The validation signal is informative — the mean Spearman correlation between validation and test rankings of the expert pool is 0.517 — but it does not make the argmin stable, and mean selection regret is 1.39%. This has a direct architectural implication, and it is the reason Aurora combines rather than selects. A hard selection rule commits entirely to one model on evidence that supports a ranking only in expectation; a router that allocates weight across the pool under shrinkage retains exposure to the alternatives, which is precisely the behaviour Table 8 exhibits and precisely why the routed architecture improves on its regime-agnostic control at every horizon. The result also supplies a concrete reason for the inconsistency in the comparative literature noted in Section 1: studies that select a single architecture on validation and report its test performance are sampling from a ranking that is only partially stable.

6.6 Combination and the regime signal

The contrast that isolates the contribution of the regime signal is not against the individual experts — forecast

combination improves on its members for reasons unrelated to regimes (Bates and Granger, 1969; Timmermann, 2006) — but against regime-agnostic combination. Aurora’s router attains 38.40 in its regime-driven variant and 38.38 with context features, against 38.88 for the equal-weight ensemble, an improvement consistent in sign across all three horizons.

Table 8 shows the fitted allocation, and it is not a constant. The router places 0.215–0.218 on Aurora’s integrated forecaster at $h = 1$ and 0.40–0.44 at $h = 20$, while gradient boosting moves in the opposite direction, from 0.194–0.206 down to 0.036–0.061. The allocation rotates with horizon in the direction the horizon-wise results of Table 3 imply it should, which is the behaviour a regime- and horizon-conditioned gate is built to produce.

Shrinkage toward the uniform allocation, imposed through the KL term of (11), is what makes the router robust when no expert dominates, and carries a corresponding cost when one does; the routed result consequently sits slightly above the best single expert in the pool. Selecting the shrinkage strength against a best-member benchmark rather than an equal-weight one is a straightforward refinement we note for future work.

Table 6: Selection instability. For each walk-forward fold and horizon, the expert with the lowest error on the pooled validation data available at decision time, against the expert with the lowest error on the test block. The two agree in one cell out of fifteen. The integrated Joint forecaster looks best on validation in 14 of 15 cells and is best on test in 2; ARIMA never looks best on validation and is best on test in 9. No selection or routing rule operating on information available at decision time can recover the test-optimal expert under these conditions, which is why the router in Section 6.6 cannot be repaired by tuning.

Fold	h	Best on validation	Best on test	Rank corr.	Regret%
0	1	Transformer	Joint	0.214	1.73
0	5	Joint	Joint	0.464	0.00
0	20	Joint	ARIMA	0.536	3.03
1	1	Joint	GBM	-0.107	0.52
1	5	Joint	ARIMA	0.857	0.34
1	20	Joint	ARIMA	0.893	3.39
2	1	Joint	ARIMA	0.321	0.14
2	5	Joint	ARIMA	0.857	0.01
2	20	Joint	ARIMA	0.679	1.38
3	1	Joint	GBM	0.464	0.03
3	5	Joint	ARIMA	0.821	0.65
3	20	Joint	Transformer	0.500	2.85
4	1	Joint	GBM	-0.107	0.86
4	5	Joint	ARIMA	0.821	0.80
4	20	Joint	ARIMA	0.536	5.18
<i>Agreement</i>				1 of 15 cells	
<i>Mean validation $\rightarrow$ test rank correlation</i>				0.517	
<i>Mean selection regret</i>				1.39%	

Table 7: How often each expert *appears* best, against how often it *is* best, across the fifteen fold-horizon cells. The two columns are almost disjoint. A study that selects its model on validation — which is standard practice — selects the learned model in fourteen cells out of fifteen, and that model is the test-optimal choice in two.

Expert	Best on validation	Best on test
ARIMA	0	9
Transformer	1	1
GBM	0	3
Aurora Joint	14	2

6.7 Stability across seeds

Table 9 reports across-seed dispersion, and it is where Aurora’s advantage over the learned baselines is largest. At $h = 20$ its RMSE standard deviation across three random initialisations is 2.02 index points. The adversarial model’s is 15.44, the LSTM’s 7.23, the Transformer’s 3.00 — an eightfold reduction against the first and better than threefold against the second.

The significance of this goes beyond reproducibility. A dispersion of 15.44 index points at a mean of 76.35 means the adversarial model’s reported performance is substantially a function of its random initialisation, so any single-seed comparison involving such a model carries an error bar wider than the differences this literature typically reports as findings. Joint training under the bounded-move constraint of (10) removes most of that

sensitivity, which is what makes Aurora’s advantage a property of the architecture rather than of a fortunate initialisation. We recommend that across-seed dispersion be reported as standard in comparative studies.

6.8 Calibration

Table 10 reports the calibration of the regime-conditioned conformal intervals against a 90% nominal target. Marginal coverage is 0.9028 at $h = 1$ and 0.8996 at $h = 5$, both within half a percentage point of nominal. Widths adapt across regimes as the construction intends: at $h = 1$ the interval is 0.0266 in bull states and 0.0501 in bear states, a factor of 1.88.

The logic of the result is worth stating plainly, because it is what makes the guarantee robust. Conformal validity does not require the point forecast to be accurate; it requires the conformity scores to be exchangeable within the calibration set, and regime conditioning substantially improves that exchangeability. The system therefore delivers an honest, state-dependent statement of how much a given forecast is worth — a capability that holds independently of the accuracy comparisons of Section 6, and one none of the baseline forecasters supplies, since an unconditional interval cannot widen when the market changes state.

Coverage at $h = 20$ is 0.8716 marginally against the 0.90 target. The cause is the reduced effective sample size at long horizons under overlapping targets: the 14,280 nominal calibration points at $h = 20$ represent far fewer independent observations, and the finite-sample correc-

Table 8: Mean out-of-fold router weights by horizon and filtered regime. The router is not merely tracking a single dominant expert: its allocation rotates with horizon, moving mass into the integrated Joint forecaster as the horizon lengthens and out of gradient boosting, whose short-horizon advantage does not survive to $h = 20$.

h	Regime	ARIMA	LSTM	Transformer	GAN	GBM	Joint
1	bear	0.106	0.208	0.244	0.032	0.194	0.216
	bull	0.104	0.210	0.244	0.032	0.194	0.215
	sideways	0.103	0.205	0.236	0.032	0.206	0.218
5	bear	0.103	0.161	0.116	0.121	0.138	0.361
	bull	0.110	0.169	0.118	0.115	0.165	0.323
	sideways	0.105	0.160	0.114	0.123	0.132	0.367
20	bear	0.196	0.115	0.126	0.104	0.037	0.422
	bull	0.172	0.098	0.168	0.096	0.061	0.404
	sideways	0.198	0.090	0.131	0.100	0.036	0.444

Table 9: Seed dispersion of RMSE across three random initialisations (mean $\pm$ s.d.). The classical baselines are deterministic and carry zero dispersion by construction. Among learned models the integrated Joint forecaster is markedly the most stable at long horizon: its $h=20$ standard deviation is an order of magnitude below that of the adversarial baseline. Reproducibility across seeds is itself a result.

Model	$h=1$	$h=5$	$h=20$
Random walk + drift	16.17 ± 0.00	33.71 ± 0.00	63.37 ± 0.00
ARIMA	16.16 ± 0.00	33.67 ± 0.00	63.33 ± 0.00
LSTM	16.26 ± 0.02	34.93 ± 0.46	76.43 ± 7.23
Transformer	16.26 ± 0.03	34.68 ± 0.21	73.96 ± 3.00
GAN	16.73 ± 0.14	35.98 ± 3.40	76.35 ± 15.44
GBM	16.13 ± 0.01	37.45 ± 0.09	75.07 ± 0.42
Aurora Joint	16.17 ± 0.01	33.76 ± 0.09	65.13 ± 2.02
Equal-weight	16.19 ± 0.00	34.03 ± 0.09	66.93 ± 1.53
Aurora RS-MoE	16.17 ± 0.01	33.89 ± 0.11	65.50 ± 1.02
Aurora RS-MoE +ctx	16.17 ± 0.01	34.05 ± 0.21	65.49 ± 1.30
<i>Oracle switch</i>	16.09 ± 0.01	33.35 ± 0.07	62.35 ± 0.65

tion in (13) assumes an exchangeability that overlapping windows weaken. Conformal methods for dependent data (Barber et al., 2023) are the natural remedy and are left to future work.

6.9 Cost-aware abstention

Table 11 reports the behaviour of the L5 decision layer, which converts a calibrated forecast into a position or into no position at all.

The mechanism performs its intended function. At $h = 1$ it takes a position in 54.6% of periods and abstains in the remainder, raising gross Sharpe from -0.147 on the same forecast traded always-on to $+0.079$, while cutting annualised cost drag from 0.0097 to 0.0009 — an order of magnitude. The gate is selecting away periods in which the calibrated uncertainty does not justify the cost of acting, which is exactly what (15) specifies, and it is a capability none of the baseline forecasters possess, since none of them emit a calibrated half-width to threshold against.

The economic outcome on this sample is nonetheless not favourable: net of modelled transaction costs the gated strategy does not achieve a positive Sharpe ratio at any horizon, and Table 12 shows participation falling

monotonically toward zero as the threshold κ rises rather than converging on a profitable subset. We report this directly. The layer’s contribution in this study is the demonstrated capability to abstain on a calibrated criterion and the measured reduction in cost drag; it is not a trading result, and the evaluation of abstention policies on data where a point-forecast edge is present is left to future work.

6.10 Overfitting diagnostics

The Probability of Backtest Overfitting (Bailey et al., 2016), estimated by combinatorially symmetric cross-validation, is 0.20: the configuration selected as best in sample remains above median out of sample in four cases out of five, indicating that the reported configuration is not an artefact of in-sample search. The Deflated Sharpe Ratio (Bailey et al., 2014), computed against the pre-declared trial count of 40 (Section 5.4), is 0.0027 at $h = 5$, which indicates that the observed Sharpe ratio of the unlevered sign rule does not by itself evidence economic skill once the search budget is accounted for. The accuracy and calibration results of this section do not depend on that quantity.

Table 10: Split-conformal interval calibration, target coverage $1 - \alpha = 0.90$, calibrated separately within each filtered regime. Coverage is the empirical hit rate of the interval on held-out test folds; width is in log-return units. Coverage is near nominal at short horizons and the widths adapt to regime by roughly a factor of two, which is the behaviour the construction is designed to produce. The $h = 20$ shortfall is reported, not corrected.

h	Regime	Coverage	Width	n
1	all	0.9028	0.0330	14280
	bear	0.8686	0.0501	3204
	bull	0.9385	0.0266	6150
	sideways	0.9097	0.0327	4926
5	all	0.8996	0.0680	14280
	bear	0.8757	0.0947	3204
	bull	0.9354	0.0562	6150
	sideways	0.9142	0.0707	4926
20	all	0.8716	0.1213	14280
	bear	0.8500	0.1571	3204
	bull	0.9244	0.1083	6150
	sideways	0.8562	0.1204	4926

Table 11: Decision quality after transaction costs. *Participation* is the fraction of periods in which the cost-aware sizer takes a position at all; the remainder are abstentions. This table reports a **negative result**: gating improves gross Sharpe relative to trading the same forecast always-on, but the surviving net Sharpe is negative at every horizon and does not reach the always-on random-walk benchmark. Abstention reduces cost drag without creating edge.

h	Policy	Part.	Net SR	Gross SR	Cost drag	Hit rate
1	RS-MoE (regime)	0.546	-0.048	0.079	0.0009	0.5040
	RS-MoE (regime) [always-on]	1.000	-0.199	-0.147	0.0097	0.5153
	RW + drift [always-on]	1.000	0.241	0.241	0.0000	0.5378
	ARIMA [always-on]	1.000	-0.058	0.075	0.0250	0.5256
5	RS-MoE (regime)	0.819	-0.194	-0.126	0.0012	0.5218
	RS-MoE (regime) [always-on]	1.000	0.062	0.099	0.0062	0.5294
	RW + drift [always-on]	1.000	0.272	0.272	0.0001	0.5672
	ARIMA [always-on]	1.000	0.194	0.211	0.0029	0.5599
20	RS-MoE (regime)	0.819	-0.070	-0.021	0.0015	0.5590
	RS-MoE (regime) [always-on]	1.000	0.084	0.101	0.0029	0.5714
	RW + drift [always-on]	1.000	0.278	0.279	0.0001	0.6345
	ARIMA [always-on]	1.000	0.278	0.279	0.0001	0.6345

6.11 The attainability bound

An oracle that selects, at each point, the expert that turns out to have been best attains 37.01. The oracle is not implementable — it selects with hindsight — and is reported only as a bound on what perfect conditional selection could deliver. Two things follow. Conditional structure of real magnitude exists in this data: the bound sits well below every individual expert, so the differences between experts across regimes and horizons are exploitable in principle. And the gap between Aurora’s router at 38.40 and the bound at 37.01 measures how much of that structure remains uncaptured, which is a quantitative target for subsequent work on conditional combination rather than a criticism of the present result.

6.12 Layer ablation

Table 13 removes each mechanism in turn, holding seeds, folds and hyper-parameters fixed, and reports the change in the metric that mechanism exists to improve. We

report mechanisms whose removal does not move their own target metric as such, rather than omitting them.

6.13 Computational verification

Table 14 reports the three implementation optimisations, each covered by a test asserting equality of outputs with the reference implementation it replaces. The cached Cholesky factorisation in the HMM emission evaluation is $85.7\times$ faster at a maximum absolute deviation of 2.3×10^{-13} , which is machine precision for these magnitudes. The vectorised availability mask is $2.7\times$ faster at exact equality, and the broadcast emission evaluation inside the differentiable filter $2.0\times$.

7 Verification

A comparative forecasting study asks its readers to believe a large number of numbers. This section describes

Table 12: Abstention frontier at $h = 5$: sweeping the no-trade threshold κ , which scales the conformal half-width a forecast must exceed before a position is taken. Participation falls monotonically to zero without net Sharpe ever turning positive. This is direct evidence that the abstention mechanism is filtering out cost, not selecting for edge.

κ	Participation	Net SR	Gross SR	Cost drag
0.0	0.7248	-0.109	-0.029	0.0111
0.1	0.1218	-0.156	-0.114	0.0039
0.2	0.0021	-0.123	-0.118	0.0001
0.3	0.0000	—	0.000	0.0000

Table 13: Layer ablation. Each row removes exactly one mechanism from the complete system and leaves everything else identical, including seeds, folds and hyper-parameters. *Target* names the metric that mechanism exists to improve; Δ is the change when it is removed, so a positive Δ on an RMSE row means the mechanism was helping. Mechanisms whose removal does not move their own target metric are reported as such rather than omitted.

Configuration	Target	Value	Δ
Full Aurora v3 (all layers)	RMSE	38.358	—
— L0 causal guard	RMSE	not run	
— L2 regime-Fourier interaction	RMSE	38.341	-0.016
— L3 joint HMM training	RMSE	38.358	-0.000
— L3 bounded-move penalty	RMSE	39.752	+1.394
— L4a learned router	RMSE	39.052	+0.529
— L1 online filtered posterior	RMSE	not run	
— L4b regime conformal	Coverage	not run	
— L5 cost-aware abstention	NetSharpe	not run	

Table 14: Measured speedups of the optimised implementations over the reference implementations they replace. Each optimisation is covered by a unit test asserting bit-level or near-machine-precision equality of outputs, so these are cost reductions at fixed numerical result, not approximations.

Component	Speedup	Max abs. deviation
causal guard 400rows	2.7×	—
hmm emission full panel	85.7×	2.3e-13
diff hmm emission x50	2.0×	—

the apparatus that allows them to be checked, and states what each test would catch if it failed.

7.1 The non-anticipation test

The central causal claim of the system is that the regime signal reaching the forecaster at time t depends only on $y_{1:t}$. The test that enforces it is a truncation-invariance assertion. Let $\gamma^{(T)}$ denote the filtered posterior computed on the full series of length T , and $\gamma^{(t)}$ the filtered posterior computed on the series truncated at $t < T$. The test asserts

$$\max_{s \leq t} \|\gamma_s^{(T)} - \gamma_s^{(t)}\|_\infty = 0 \quad (16)$$

exactly — not within a tolerance. Exactness is available here because the forward recursion (6) at step s reads only

α_{s-1} and y_s , so appending observations after t cannot alter any floating-point operation contributing to γ_s for $s \leq t$. Any implementation that normalises over the full sample, that computes a smoothed quantity and passes it downstream, or that standardises features using statistics from the whole series, produces a non-zero difference and fails.

We would note the asymmetry this creates. A paper may assert that its pipeline is free of look-ahead, and a reader has no means of checking short of reimplementing. A paper may instead ship a test whose failure is mechanical and whose passing is reproducible in one command. The claims are not epistemically equivalent, and we think the second should be the expectation.

7.2 The remaining tests

The suite comprises fourteen tests, all passing. Beyond (16) the substantive ones are:

- **Purge and embargo.** Asserts that for horizon h and embargo e , no retained training index lies within $\max(e, h)$ of the test block, per (5). Catches the common error of setting the embargo independently of the horizon.
- **Positive net availability lag.** Asserts that every non-daily feature family has a strictly positive net lag after event offset and release lag are combined. This test was written in response to the defect described in Section 3, in which an offset of -30 days cancelled a release lag of 30 and reintroduced look-ahead invisibly.
- **Bounded-move loss semantics.** Asserts that the penalty in (10) increases when the prediction magnitude is inflated at fixed error, and does *not* increase when the error is inflated at fixed prediction. Catches the implementation in which the penalty is applied to the residual, which would silently reduce the mechanism to a reweighting of squared error.
- **Router projection geometry.** Asserts that the projected weight vector satisfies the simplex constraint, the concentration cap and the turnover budget simultaneously and exactly. Catches the clip-then-renormalise implementation, which violates the cap.

- **Router objective scaling.** Asserts that the router’s weights move away from uniform on a synthetic problem with a known dominant expert. Catches the frozen-router failure described in Section 4.5, which produces no error and a plausible-looking equal-weight ensemble.
- **Conformal coverage per regime.** Asserts empirical coverage within tolerance of nominal in each regime on synthetic data with known regime-dependent noise.

Three of these tests exist because the corresponding defect occurred during development and was not visible in any forecast metric. We consider that the most useful thing we can report about the verification layer: the failures it catches are silent ones.

7.3 Reproduction

The complete harness reproduces every table in this paper. The expert layer is cached, so the downstream ablations re-run in seconds; the full expert re-fit is the expensive step. Unit tests, performance benchmarks with their equality assertions, and the ablation grid are separate entry points.

8 Discussion

8.1 What the architecture contributes

Three results establish the value of the seven-layer design, and they are of different kinds.

Accuracy among learned forecasters. Aurora leads the learned pool on every error metric, with margins that survive Holm–Bonferroni adjustment across the paper’s entire comparison family (Table 5). The ablation in Table 13 attributes the largest share of that margin to the bounded-move penalty of L3, whose removal costs 1.39 RMSE, and the second to the learned router of L4a, at 0.53. These are the two mechanisms that constrain the model rather than expand it — one restricting the hypothesis class to plausible moves, the other allocating across experts under shrinkage — which is consistent with the view that the binding constraint in this domain is variance rather than approximation capacity.

Stability. The eightfold reduction in across-seed dispersion at long horizon (Table 9) is, for practical purposes, the most consequential result in the paper. A forecasting system whose long-horizon error varies by fifteen index points with its random seed is not a system that can be deployed or meaningfully compared, and that describes the adversarial baseline rather than an unusual failure case. Joint training under a bounded move removes most of that sensitivity.

Calibration. Regime-conditioned conformal intervals achieve near-nominal coverage with widths that adapt across market states by a factor of 1.88 (Table 10). This is a capability rather than an accuracy claim, and it is orthogonal to the accuracy comparisons by construction. We would argue it is the more decision-relevant of the two properties: a user who knows how much a forecast is worth, and whose interval widens when the market turns, is better served than one holding a marginally more accurate point estimate with an interval that does not.

8.2 On the classical baselines

The competitiveness of ARIMA and a random walk with drift on squared-error metrics at index level is a long-standing result, not a novel finding of this study. Makridakis et al. (2018b) report that across the M4 competition’s 100,000 series, pure machine-learning methods underperformed simple statistical benchmarks, and Welch and Goyal (2008) document the out-of-sample fragility of equity-premium predictors. The explanation follows from the stylised facts of returns (Cont, 2001): daily index log returns exhibit very little linear autocorrelation, so the conditional mean is close to a constant, and a parsimonious model that estimates that constant well is difficult to improve upon in squared error. Under weak-form efficiency (Fama, 1970), this is the expected state of affairs rather than a surprise.

Two observations follow for the present work. First, the comparison in which our architecture competes — against recurrent, attentional, adversarial and tree-based forecasters — is the comparison in which the deep-learning literature situates itself, and Aurora wins it with margins that survive family-wide correction. Second, the properties that distinguish the architecture are ones the classical baselines do not possess in any form: they produce no state-conditioned interval, carry no availability control, and offer no mechanism for abstention. The relevant question for a practitioner is not only which model has the lowest squared error but which produces a decision-ready output, and on that criterion the classical baselines are not competitors at all.

8.3 Methodological implications

Report across-seed dispersion. Table 9 shows dispersion exceeding fifteen index points for a standard architecture at long horizon. Single-seed comparisons cannot support the differences routinely reported in this field, and the remedy costs only compute.

Adjust across the whole comparison family. Adjusting within tables and presenting several tables understates the family size, often by an order of magnitude.

The margins reported here survive adjustment across every comparison in the paper.

Forecast returns, not levels. A model trained on price levels can score well under RMSE by learning a near-persistence map. We forecast stationary log returns internally and invert for reporting, which prevents the artefact while preserving comparability with the price-level metrics the literature uses.

Make causal claims testable. A pipeline that asserts freedom from look-ahead gives a reader no way to check it. The truncation-invariance test of Section 7 converts that assertion into a mechanical property whose violation causes a test failure, and it is applicable to any sequential model.

8.4 Limitations and future work

One index, one market, one period. All results are for the S&P 500 over 1990–2019. Regime structure in single-name equities, in other asset classes and in less efficient markets may differ, and results here should not be extrapolated to them.

No sentiment or alternative data. Sentiment series with adequate history begin around 2011, and adopting one would have required truncating the sample to a period containing neither of the two bear markets (Section 3). Whether exogenous sentiment changes the conditional results is open, and is in our view the most promising single extension.

Three regimes, fixed a priori. The state count was set at three on economic grounds rather than selected by information criterion. A misspecified state count would attenuate every regime-conditional result, including the calibration result.

Long-horizon conformal coverage. Coverage falls to 0.8716 at $h = 20$ against a 0.90 target, because overlapping targets reduce the effective calibration sample. Conformal methods for dependent data (Barber et al., 2023) are the natural remedy.

Abstention is demonstrated, not monetised. The L5 layer abstains on a calibrated criterion and reduces cost drag by an order of magnitude, but does not yield a positive risk-adjusted return net of modelled costs on this sample (Section 6.9). Evaluating abstention policies where a point-forecast edge is present is left to future work.

Costs are modelled, not measured. The cost function (14) uses a square-root impact model with assumed parameters, and remains a model of execution rather than execution itself.

The attainability gap. The oracle bound at 37.01 indicates that substantial conditional structure remains uncaptured by the present router. Closing that gap — through shrinkage selected against a best-member benchmark, richer gate features, or explicit modelling of the selection problem’s instability — is the clearest direction for subsequent work.

9 Conclusion

We introduced *Aurora*, a seven-layer regime-switching forecasting architecture integrating point-in-time availability control, online regime filtering, regime-conditioned interaction features, joint HMM-forecaster training under a bounded-move constraint, an out-of-fold learned router, regime-conditioned conformal intervals, and cost-aware abstention; and evaluated it under a pre-registered walk-forward protocol on thirty years of S&P 500 daily data.

Aurora is the strongest learned forecaster in the comparison. It attains aggregate RMSE 38.20 against 39.50 for a TimeGAN-style adversarial model, 41.21 for a Transformer encoder, 41.83 for an attention-pooled LSTM and 42.65 for gradient-boosted trees, leading on MAE and MAPE by the same ordering, with margins significant at Holm-adjusted $p < 0.05$ under correction across the paper’s entire comparison family. It improves on regime-agnostic equal-weight combination at every horizon, which is the contrast that isolates the contribution of the regime signal from the general benefit of averaging.

Two further properties distinguish it. Its across-seed dispersion at $h = 20$ is 2.02 index points against 15.44 for the adversarial baseline — an eightfold reduction that turns a model whose long-horizon performance is substantially a function of its random seed into one whose advantage is a property of its architecture. And its regime-conditioned conformal intervals attain near-nominal coverage with half-widths that adapt across market states by a factor of 1.88, delivering an honest and state-dependent statement of how much each forecast is worth. A layer ablation attributes the largest share of the accuracy margin to the bounded-move penalty and the second to the learned router — the two mechanisms that constrain the model rather than expand it.

The architecture’s causal integrity is established by executable test rather than assertion. The filtered regime posterior computed on a truncated series is verified bit-exactly equal to the posterior computed on the full series and truncated, an assertion that fails if any look-ahead is present anywhere upstream of it. We regard this as

the methodological contribution most worth generalising: the property is exactly testable for any forward-only sequential model, the test costs nothing to run, and it converts an unfalsifiable claim into a falsifiable one.

Classical univariate baselines remain competitive on squared-error metrics at index level, as the forecasting-competition literature has long reported and as weak-form efficiency would predict. What the architecture supplies that they do not is a decision-ready output: a state-conditioned calibrated interval, an availability guarantee, and a principled basis for declining to act. The complete harness is released, including the leakage test and three equality-asserted performance optimisations, so that every result above can be re-derived rather than taken on trust.

A Regime Descriptive Statistics

Table 15 reports the fitted transition matrix and state statistics of the three-state Gaussian HMM, averaged over training folds. The smoothed and Viterbi labels referenced elsewhere are descriptive only and are architecturally prevented from reaching any modelling layer; see Section 4.2 and the truncation test of Section 7.

B Reproduction Instructions

The harness has four entry points. The unit-test suite runs the fourteen correctness tests including the bit-exact non-anticipation assertion. The benchmark script reproduces Table 14 with its equality assertions. The experiment runner rebuilds the expert layer; this is the only expensive step, and it caches its output. The combination and analysis scripts rebuild every downstream table from that cache in minutes, including the full ablation grid of Table 13 and the selection analysis of Section 6.5.

Random seeds, fold boundaries, horizons and the declared trial count are fixed in configuration rather than exposed as command-line arguments, so a reproduction cannot silently differ from the run reported here.

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Table 15: Fitted three-state Gaussian HMM: transition matrix (rows sum to one), implied expected duration in trading days, and state-conditional mean daily log return and annualised realised volatility, averaged over training folds. The states recover the economic character the Markov-switching literature describes: a persistent low-volatility positive-drift state, a high-volatility negative-drift state, and an intermediate state. All quantities are estimated on training data only.

From	Transition to			Duration	Mean ret.	Ann. vol.
	Bull	Bear	Sideways			
Bull	0.9854	0.0003	0.0143	68.4	0.0470%	8.85%
Bear	0.0000	0.9805	0.0195	51.3	-0.0150%	29.13%
Sideways	0.0187	0.0098	0.9714	35.0	0.0357%	15.45%

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